RECOMMENDED HYDRAULICS PRESSURE MODEL

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Chapter 1: Introduction

A primary function of a hydraulics engine is to calculate pressure during wellbore operations. The formation exerts pressure on the wellbore, which is counteracted by the drilling fluid that circulates through the wellbore. Maintaining the appropriate wellbore pressure is essential for any wellbore operation.

The Secure Hydraulics Modeling project's goal was to create a numerical framework that replicated the calculations used by Secure's current hydraulics engine, with comparable numerical results. Development for an isothermal pressure model has been completed, and additional pressure models are currently being developed.

1.1 Wellbore Conditions

One critical function of a drilling fluid is to maintain the wellbore stability. If there isn't enough pressure, formation fluids can flow into the wellbore. If there is too much pressure, the formation can fracture and drilling fluids can flow into the formation. To maintain this delicate balance, accurate pressure calculations must consider many factors that are unique to the wellbore environment and to each wellbore operation.

1.1.1 Geometry

The primary components of the wellbore are the drillstring and the annulus. The drillstring consists of drillpipe sections, the bottomhole assembly (BHA), and other tools. Drillpipe sections are connected with tool joints, which are welded onto the ends of the drillpipe. The BHA is located at the end of the drillstring. It provides the hydraulic force to rotate the bit and contains special tools, such as the drill bit, stabilizers, drill collars, measurement while drilling tools (MWDs), and mud motors. At the end of the BHA, the drill bit cuts through the formation and shoots fluid out at high speeds.

The space between the formation and the drillstring is the annulus. The annulus may be exposed to the formation or lined with casing, sections of steel pipe that are cemented into place. The annulus contains a blowout preventer (BOP), which can use a choke to control wellbore pressure if formation fluids enter the wellbore due to insufficient pressure.

1.1.2 Fluid

Drilling fluid contains salts, low-gravity solids and a base fluid. The base fluid may be oil, water, or synthetic, depending on the formation characteristics. Density and rheology are the most important fluid properties for calculating isothermal pressure loss. Fluid density, also known as mud weight, is the mass of the fluid within a specific unit of volume.

Rheology describes how shear stresses affect a fluid's shape and flow. Shear stress is caused by two objects sliding against one another, such as when a fluid flows through a pipe. Fluid flow behavior is dependent on viscosity, shear rate, and yield stress.

- *Viscosity* measures the fluid's resistance to flow. Highly viscous fluids, such as syrup, deform more slowly and require more force to move the fluid.

- The *shear rate* is the rate at which a shear force is applied, and is related to the rate at which fluid velocity changes across the pipe. At the pipe wall, the velocity is assumed to be zero (no-slip condition) and increases as the distance from the pipe wall increases. Maximum velocity occurs in the center of the pipe.
- Some fluids have a *yield stress*, which is the minimum shear stress required for fluid flow. Once the fluid at the pipe wall has enough shear stress to exceed this value, the fluid will begin to flow. In laminar flow, the yield stress is not exceeded near the center of the pipe, creating a plugged fluid region that is carried by the fluid at the pipe wall. The fluid in this plugged region flows at maximum velocity.

1.1.3 Wellbore Operations

The hydraulics engine simulates circulation, drilling, and tripping operations. During circulation, fluid is pumped from mud pits on the surface to the standpipe, then to the drillstring. Fluid flows down the drillstring, through the BHA, and across the bit. When the fluid travels across the bit, it picks up the cuttings that are generated when the bit drills through the formation. The fluid returns to the surface through the annulus, where the cuttings are removed before the fluid is re-circulated through the system.

Drilling and circulation are often performed simultaneously. During a drilling operation, the bit sits on the bottom of the borehole and cuts into the formation while the drillstring rotates. The cuttings generated by the bit are carried to the surface by the fluid. The depth of the wellbore increases, which increases the fluid required to fill the wellbore.

Tripping operations may be performed with or without circulation. A trip is the process of moving the drillstring into or out of the wellbore to make connections at the surface. During this process, pipe is added or removed at the surface.

- Pipe is pushed into the wellbore during a trip *in*. The space in the wellbore that is occupied by pipe steel increases, which decreases the total fluid volume in the wellbore. The additional pipe steel displaces some of the fluid, and the displaced volume exits the wellbore through the annulus.
- Pipe is removed from the wellbore when tripping *out*. The space occupied by pipe steel decreases, which increases the total fluid volume in the wellbore. The drilling fluid moves to the bottom of the wellbore from the pipe and annulus to fill the space vacated by the pipe steel. Depending on the amount of pipe removed from the wellbore, fluid may be added to the drillstring from the trip tank.

1.2 Pressure Loss

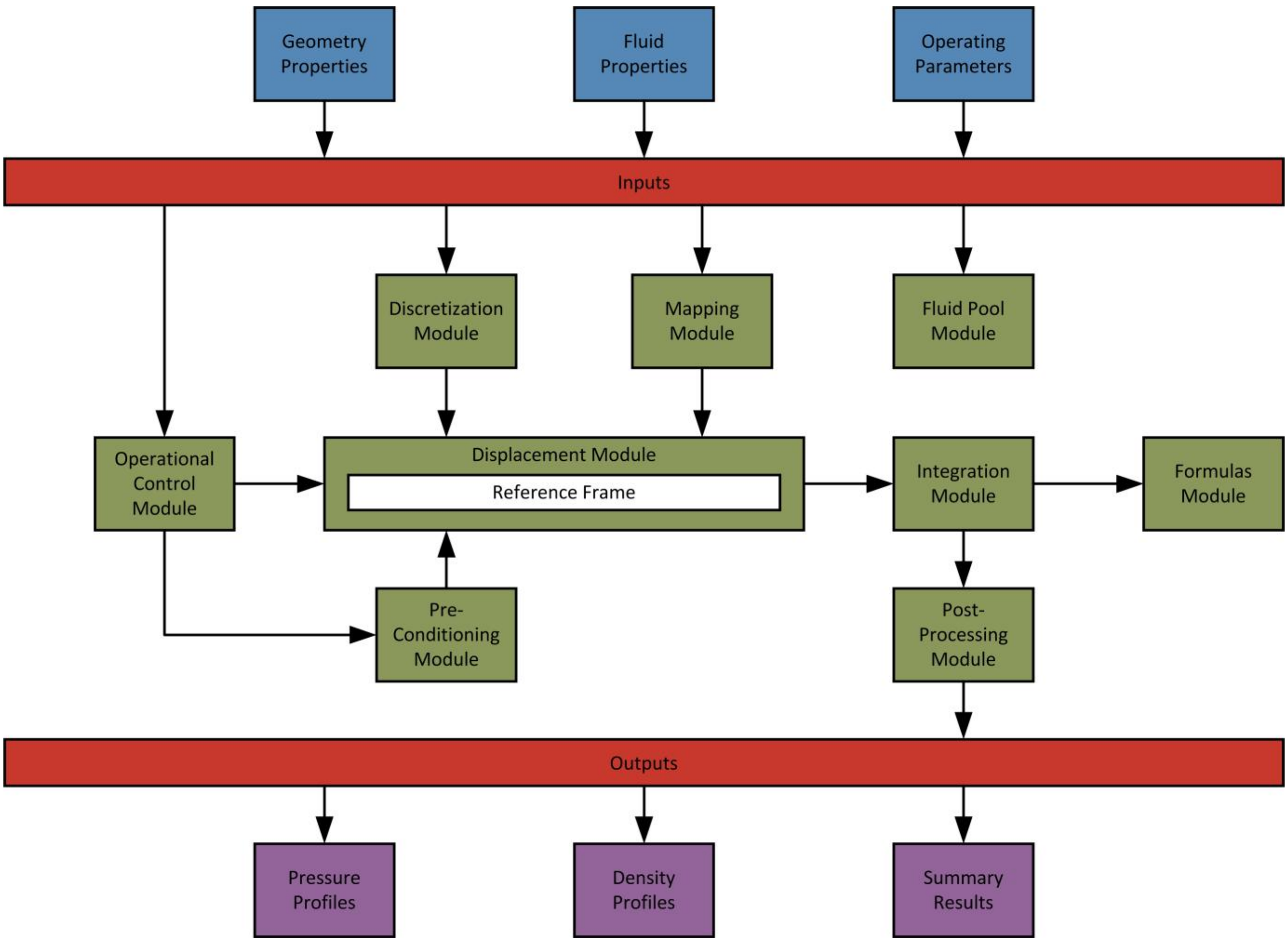
The formation surrounding the wellbore also contains fluids, known as formation fluids, which exert pressure on the wellbore. If the formation pressure is greater than the pressure exerted by the drilling fluid, formation fluids will flow into the wellbore. However, if the drilling fluid's pressure is too high, the formation will fracture, drilling fluid will flow into the formation, and annular sections of the wellbore may collapse. Therefore, the fluid's pressure should be greater than the formation pressure, but less than the formation's fracture gradient pressure.

The fluid exerts hydrostatic and frictional pressure. Hydrostatic pressure is caused by the effects of gravity on the drilling fluid's mass. It is primarily affected by the fluid density and true vertical depth (TVD) at a particular point in the well. Hydrostatic pressure increases as the depth and density of the drilling fluid increases.

Frictional pressure loss results from the friction created by fluid molecules rubbing against the walls of the pipe and annulus. In order for the fluid to circulate, the mud pumps must have sufficient hydraulic power to overcome the pressure losses throughout the system. Factors such as viscosity, fluid location, and flow regime significantly influence frictional pressure loss. Additional pressure loss is caused by geometry changes, pipe rotation, flow through the nozzles of the bit, and swab and surge due to tripping operations.

Chapter 2: Simulation Process

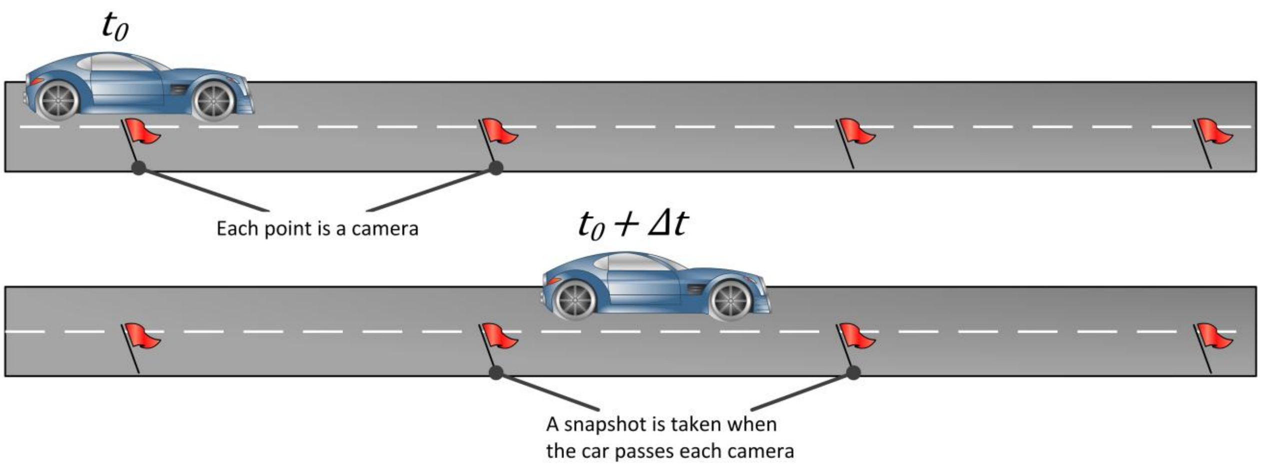
Within the hydraulics engine, the numerical framework uses process modules and a reference frame to simulate wellbore operations. The modules store data, coordinate simulations, and perform calculations. The reference frame is a coordinate system that uses a specially constructed data structure to represent the fluid flowing in the wellbore. The wellbore is divided into cells, using a process known as discretization. The reference frame holds these cells and uses them to track the fluid as it travels through the wellbore. Discretization also makes the calculations performed by the process modules more manageable.



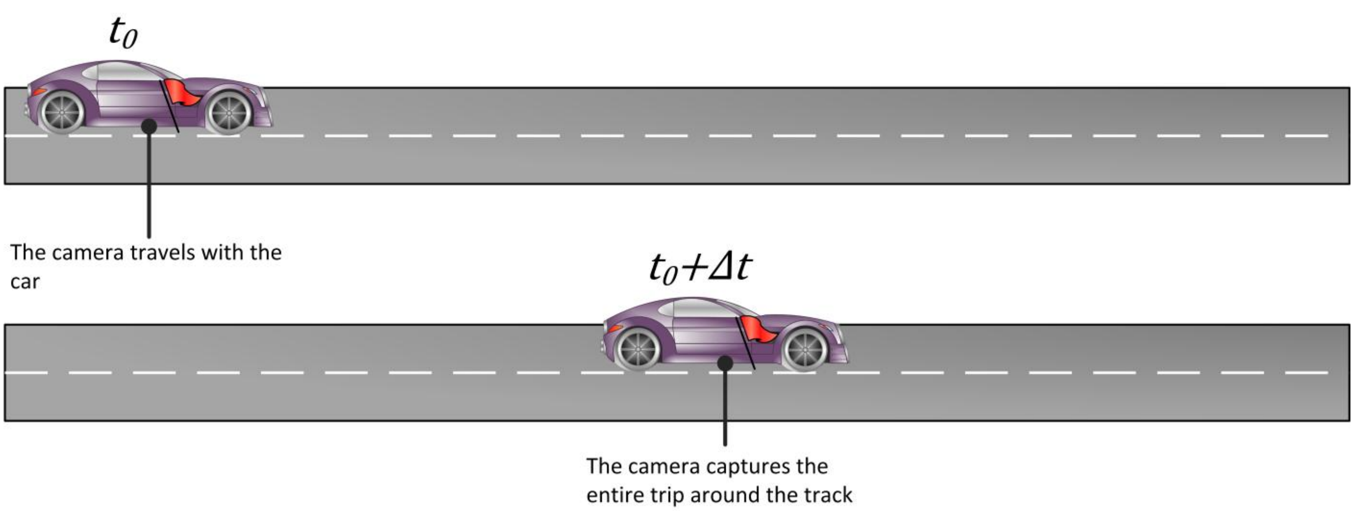
2.1 Reference Frame

The numerical framework uses a Lagrangian reference frame, rather than a traditional Eulerian reference frame, to represent the fluid in the wellbore. In a Lagrangian reference frame, the discretized cells move through the wellbore with the fluid. Each cell represents a specific segment of the drilling fluid and carries the properties of that fluid segment with it. This is different than an Eulerian reference frame, which uses fixed points to capture fluid properties.

Imagine a car driving around a race track. Cameras are placed ten feet apart around the track. When the car passes a camera, the camera records the car's passing. At the end of the race, the footage from the cameras is collected and analyzed. The snapshots taken of the car give a good approximation of the car's trip around the race track. However, it isn't a complete picture of the car's behavior because there is no record of the spaces between each camera.



When a second car drives around the track, a single camera is placed inside the car. The camera travels with the car and captures the car's entire trip around the race track. The resulting footage is an exact record of the car's entire trip around the race track. Because the camera traveled with the car, there are no gaps in the data.



The cameras represent observers collecting data for each car, and the race track represents the fluid's path. The first car uses an Eulerian reference frame, where an observer collects data at fixed points along the fluid's path.

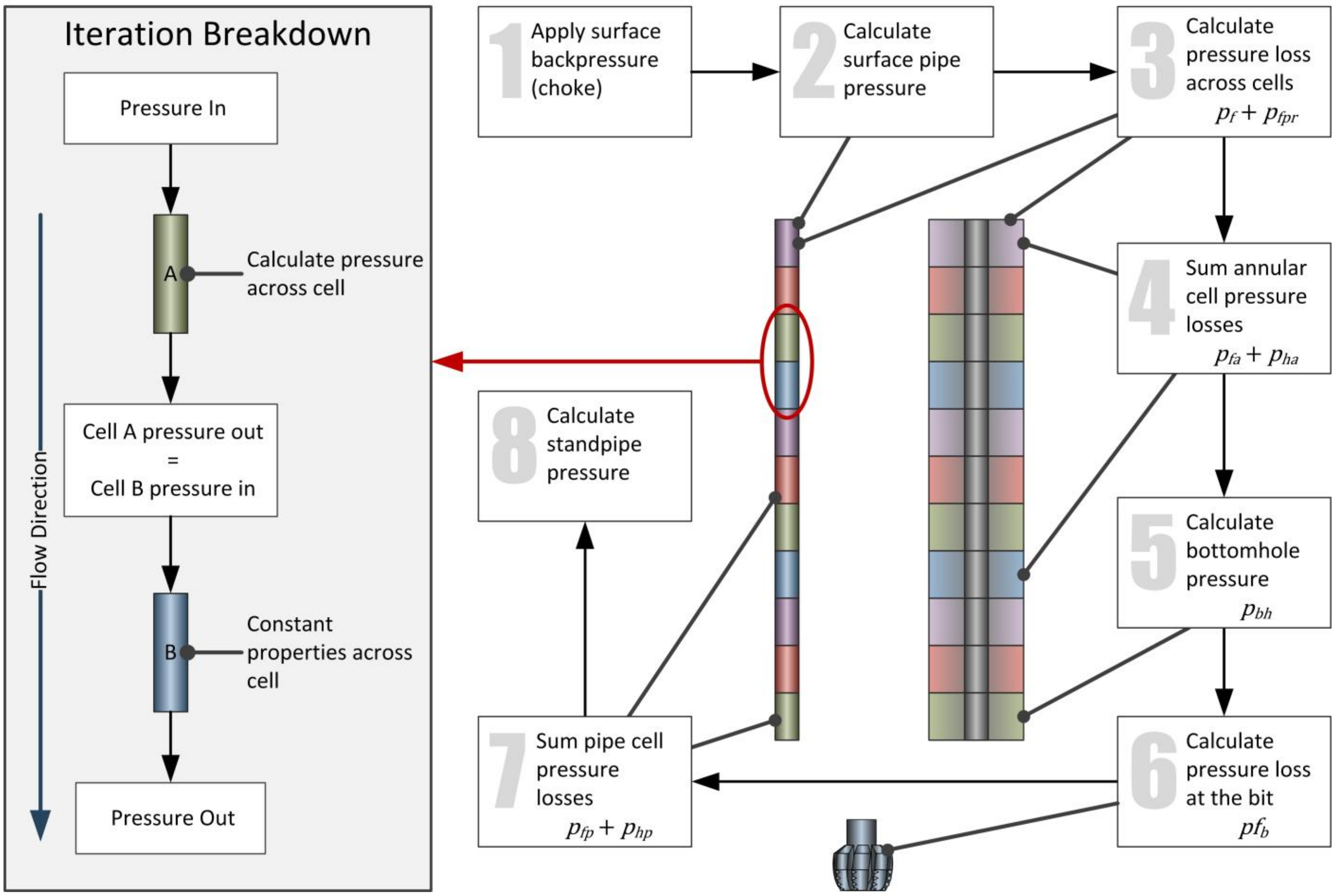
The second car uses a Lagrangian reference frame, where the observer moves with the fluid to collect data. In the example above, the camera placed inside the second car represents the observer.

The Lagrangian reference frame provides many advantages over an Eulerian reference frame. It allows the fluid to be tracked as it moves through the wellbore, which is not possible with an Eulerian reference frame, and eliminates the need to re-discretize the wellbore when the drillstring moves. Calculations for hole cleaning and thermal circulation are also simplified by a Lagrangian reference frame.

In most flow simulations, velocity is an unknown variable. The Eulerian reference frame is more common because the Lagrangian reference frame is dependent on the flow velocity. Since the flow rate at the surface is known in most wellbore simulations, average velocity can be calculated from the flow rate when the flow is incompressible and one-dimensional.

The Lagrangian coordinate system also responds easily to drillstring movement. In a Eulerian system, geometric properties are stored in the cells when the wellbore is discretized. When the drillstring moves, the wellbore has to be re-discretized. In the Lagrangian reference frame, however, the geometric properties of each cell are mapped at each timestep. When the drillstring moves, cell positions are adjusted without re-discretizing the wellbore.

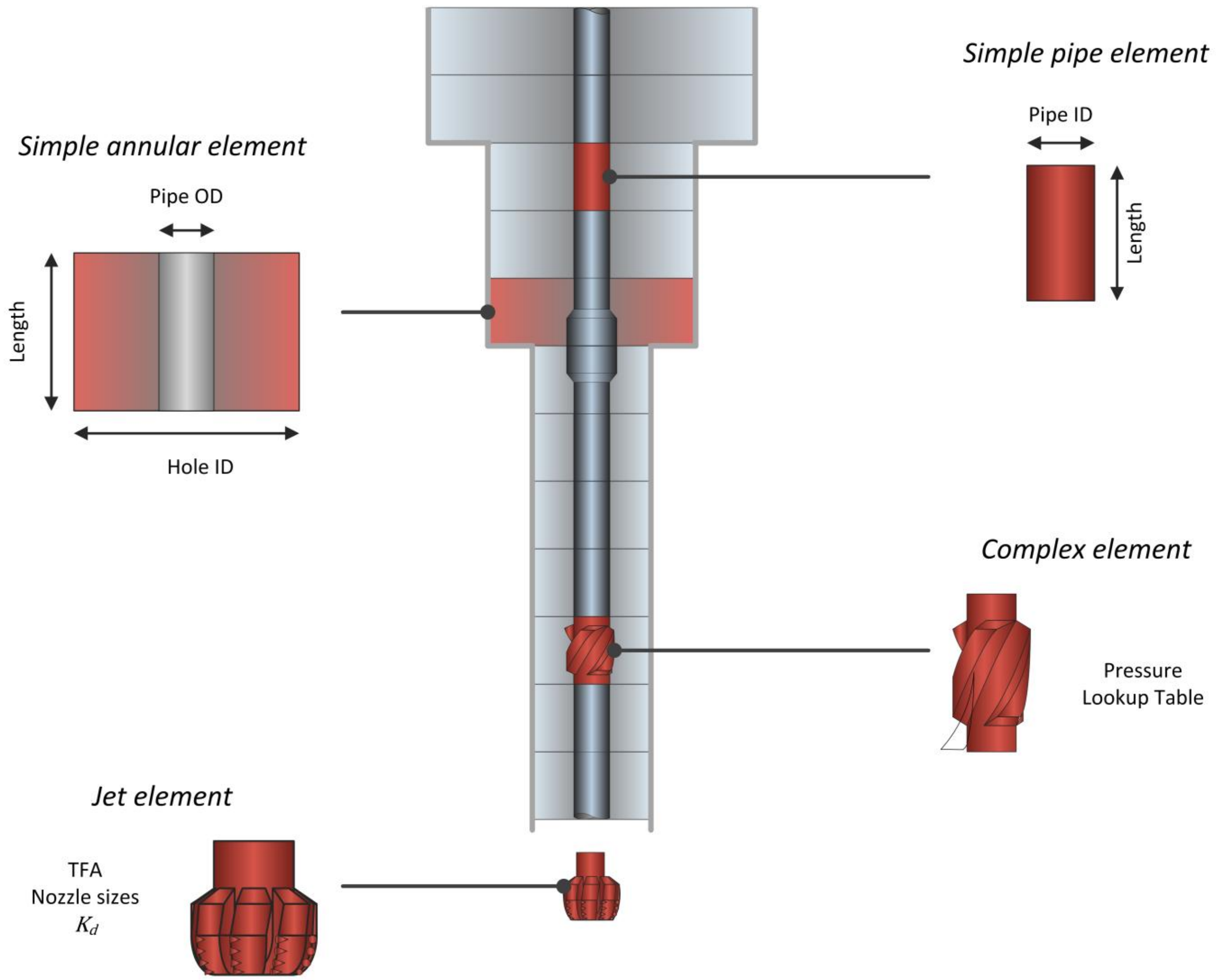
Pressure is calculated for each cell by using geometric and fluid parameters. Total pressure is determined from the sum of the individual cell pressures. For isothermal pressure calculations, cell pressure can be integrated in any order. However, pressure must be calculated in the reverse direction of fluid flow - from the annulus to the drillstring - for thermal pressure calculations.



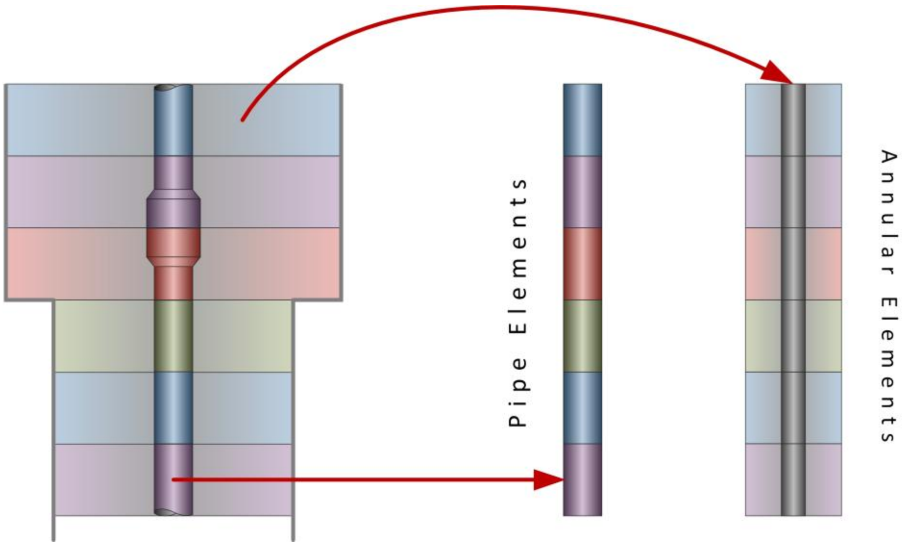
2.2 Discretization

There are thousands of possible wellbore components, but they can be classified as simple, jet, or complex elements. Simple elements are pipe and annular sections, and they are used to represent most of the wellbore. Properties such as outside diameter, inside diameter, and length are used to characterize simple elements.

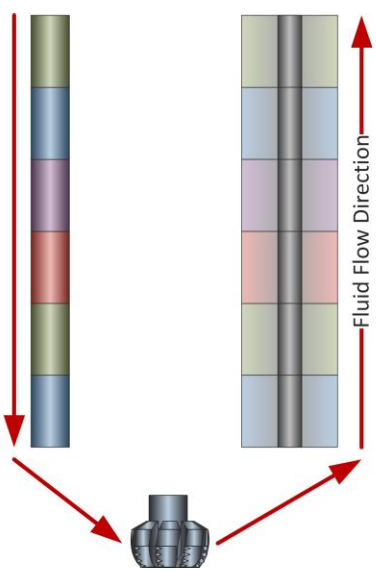
Jet elements, such as the bit, contain nozzles or ports. Flow is diverted inside these elements when fluid flows through a bit or out of a port. Jet cells are characterized by properties such as total flow area, nozzle sizes, and efficiency. Complex elements represent regions of the wellbore, such as stabilizers and MWD tools, that cannot be classified as simple or jet elements. They have irregular geometry, which creates intricate flow paths. These elements require mathematical models, which will be included in a later development phase.



Most of the cells represent simple elements, each holding a specific segment of fluid volume. They contain the geometric and fluid properties for that segment of the drilling fluid. Although cell properties may change as the cell moves through the wellbore, these properties are assumed to be constant across the cell.



Cells fill the wellbore's entire fluid volume without overlapping one another. They are connected end-to-end in the order they are encountered in the wellbore, creating a list which shows their progression along the fluid path. The first cell on the list represents the fluid at the top of the drillstring and the last cell represents the top of the annulus. As fluid is pumped into the wellbore, new cells are added at the top of drillstring. Cells are pushed through the wellbore by incoming fluid, giving each cell a new position without changing its volume. In isothermal simulations, the volume added to the system by the new cells is removed by the cells exiting the system through the annulus. In thermal simulations, the amount of fluid volume removed from the annulus is determined by the mass of the fluid pumped into the system.



2.3 Processes & Modules

The numerical framework uses management processes and modules to create outputs. The management processes oversee the modules to ensure each task associated with the process is completed.

The initialization process creates an initial snapshot of the wellbore that serves as the starting point for the simulation. Cells are constructed to represent the wellbore, then filled with local properties. The resulting system reflects the initial conditions of wellbore simulation.

Once the initialization process is completed, the iteration process can be started. During this process, the simulation advances one timestep. Cells are adjusted to reflect the resulting changes and the calculated properties are communicated to the next timestep.

2.3.1 Discretization Module

The initialization process stores geometric properties in the discretization module, which uses these properties to split the wellbore into cells. The cells represent a segment of fluid volume in the wellbore and have local geometric, fluid, operational and calculated properties attached to them. Cell volume is used to track the cell as it moves through the wellbore. The cumulative volume of the system, which is used by the mapping module, is also tracked by the discretization module.

2.3.2 Mapping Module

The mapping module creates a geometric map of the wellbore that relates cumulative volume to the measured depth of each cell. It also creates a survey map of the wellbore's trajectory that relates the measured depth along the curve of the wellbore to the total vertical depth.

Using these maps, geometric and survey properties are assigned to cells after the wellbore is discretized. Cell position is determined by the geometric map, based on the cell's volume. The mapping module assigns the cell's geometric and survey properties by cross-referencing cell position with the cumulative volume data stored in the discretization module. When cells move during an iteration, the mapping module updates the cells with their new geometric and survey properties.

2.3.3 Fluid Pool Module

The fluid pool module stores the fluid data and fluid-related calculations. The initialization process populates the fluid pool module with properties such as density, rheology, and PVT data. When the mapping module determines a cell's position, these properties are mapped to each cell.

2.3.4 Displacement Module

The reference frame, which contains the discretized cells, resides in the displacement module, which manipulates them as instructed by the discretization, pre-conditioning, and operational control modules. The displacement module adds and removes cells from the system, adjusts cell positions, and merges and splits cells.

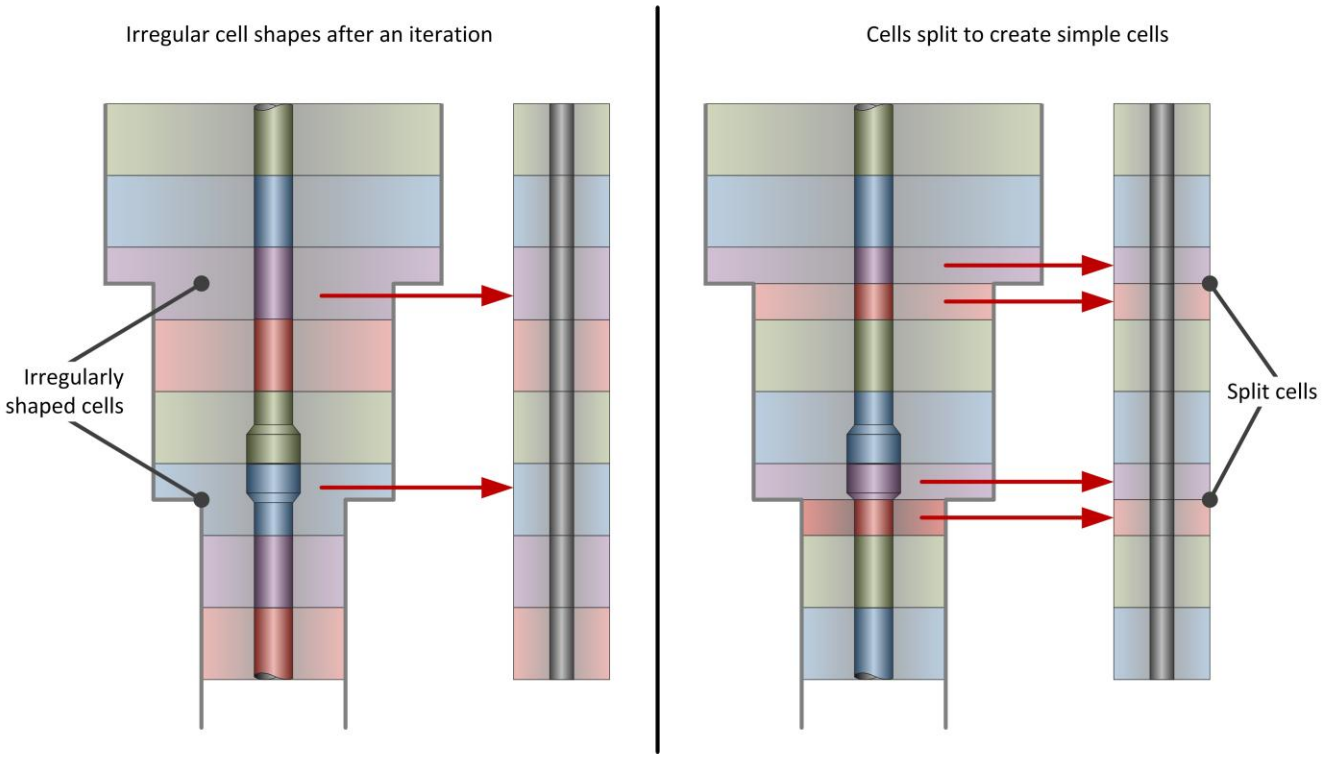
When the displacement module splits a cell, it creates two or more new cells from an existing cell. The new cells have the same properties as the original cell. The sum of the volume of the new cells is the same as the volume of the original cell. Splitting cells can resolve problems with the geometry mapping and the wellbore discretization refinement process.

When two cells are merged, a new cell is created to replace multiple cells. The volume of the new cell is equal to the combined volumes of the merged cells. When cells are merged, their properties are combined using a mathematical model.

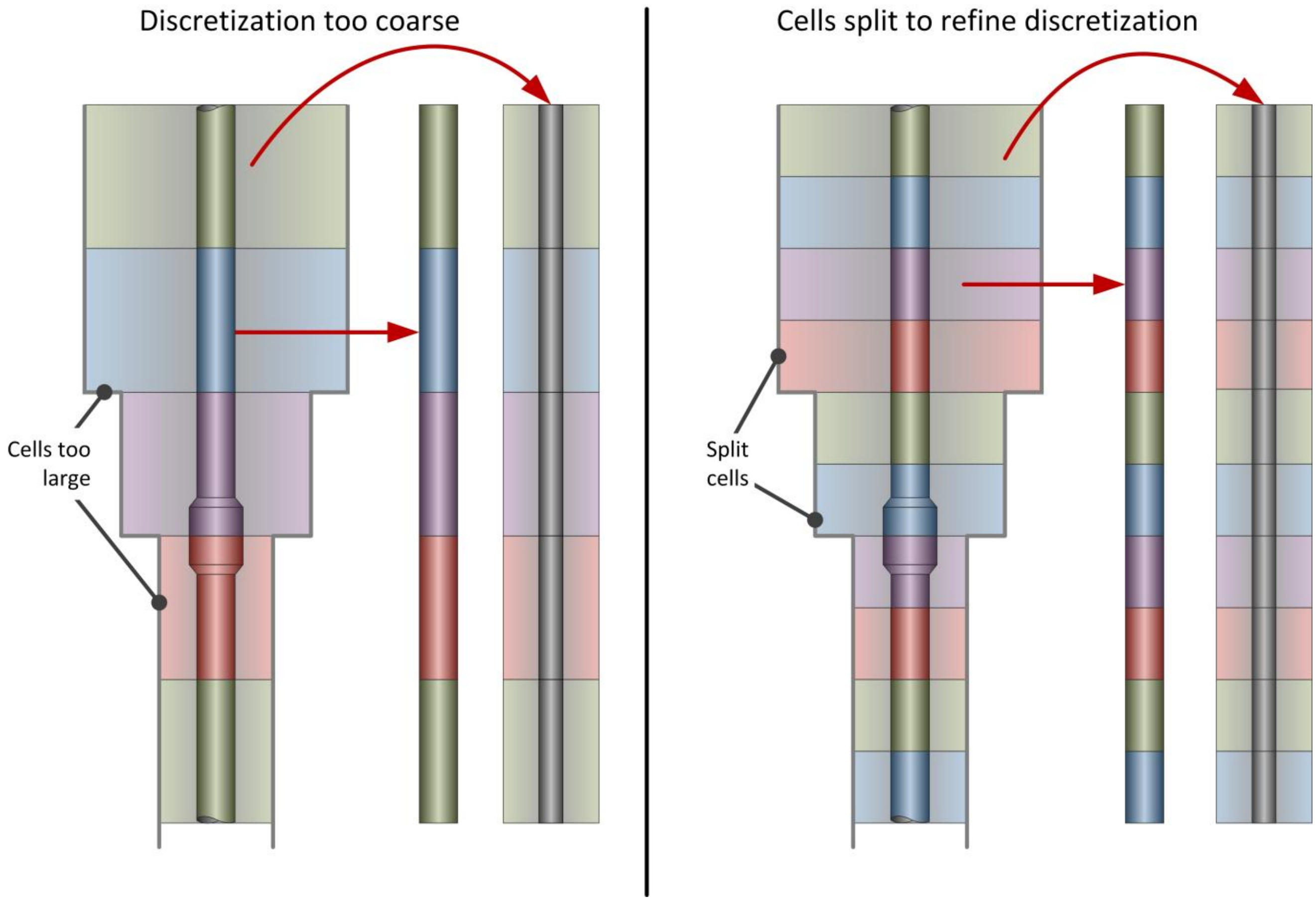
2.3.5 Pre-Conditioning Module

The pre-conditioning module communicates with the displacement module to optimize cell distribution and the number of cells. It tells the displacement module how to merge and split the cells, which can optimize performance by finding a balance between speed and quality.

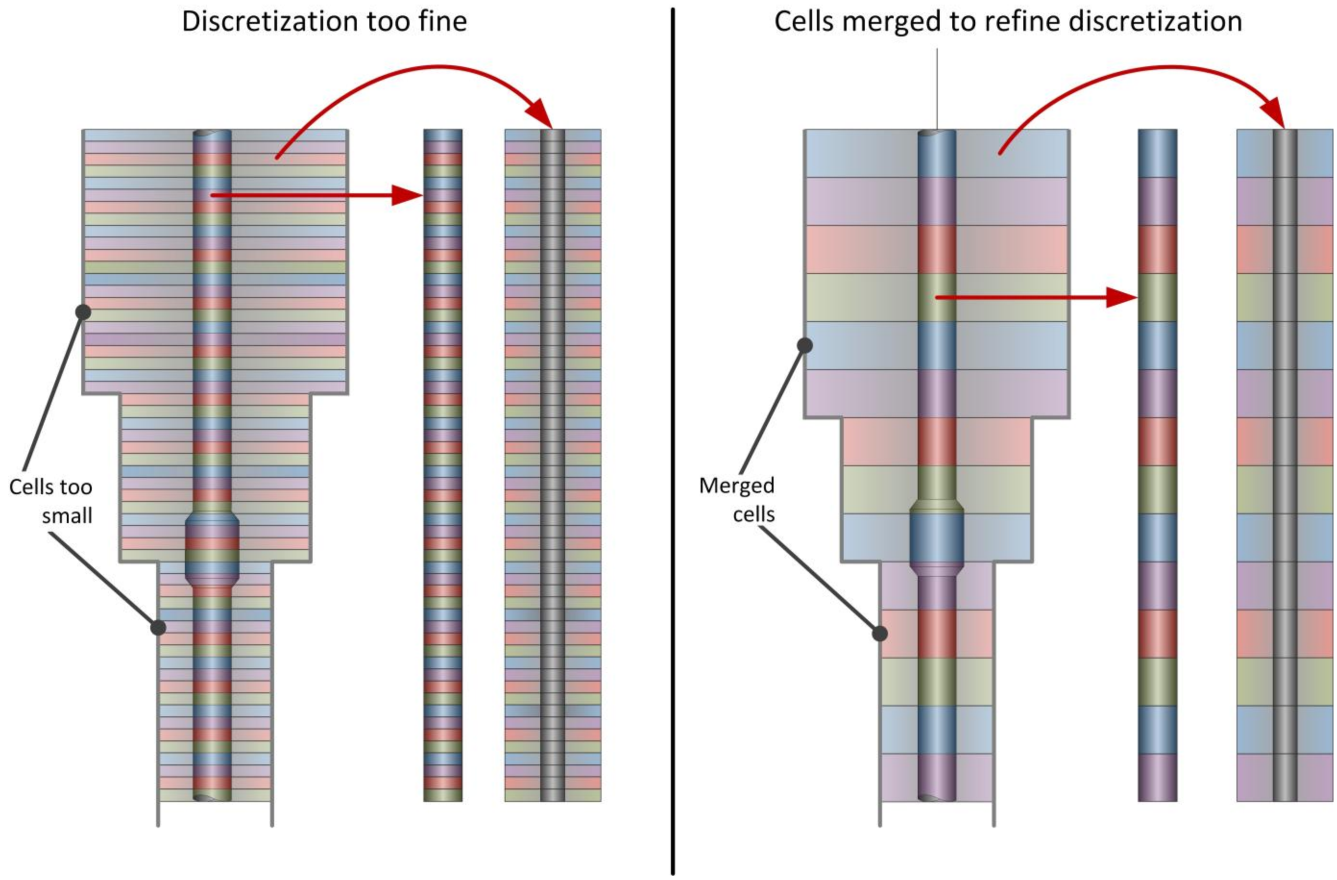
After an iteration, some simple cells may have an irregular shape. This can happen when a cell crosses a geometry change, such as the boundary between two casing intervals. This problem is resolved by splitting the cell into simple cells along the geometry change.



If cells are too large, splitting cells may be required. Large cells indicate that there may not be enough cells in the wellbore, causing poor integration quality and unstable calculations. Splitting cells can refine cell distribution and improve integration quality, which leads to better accuracy.



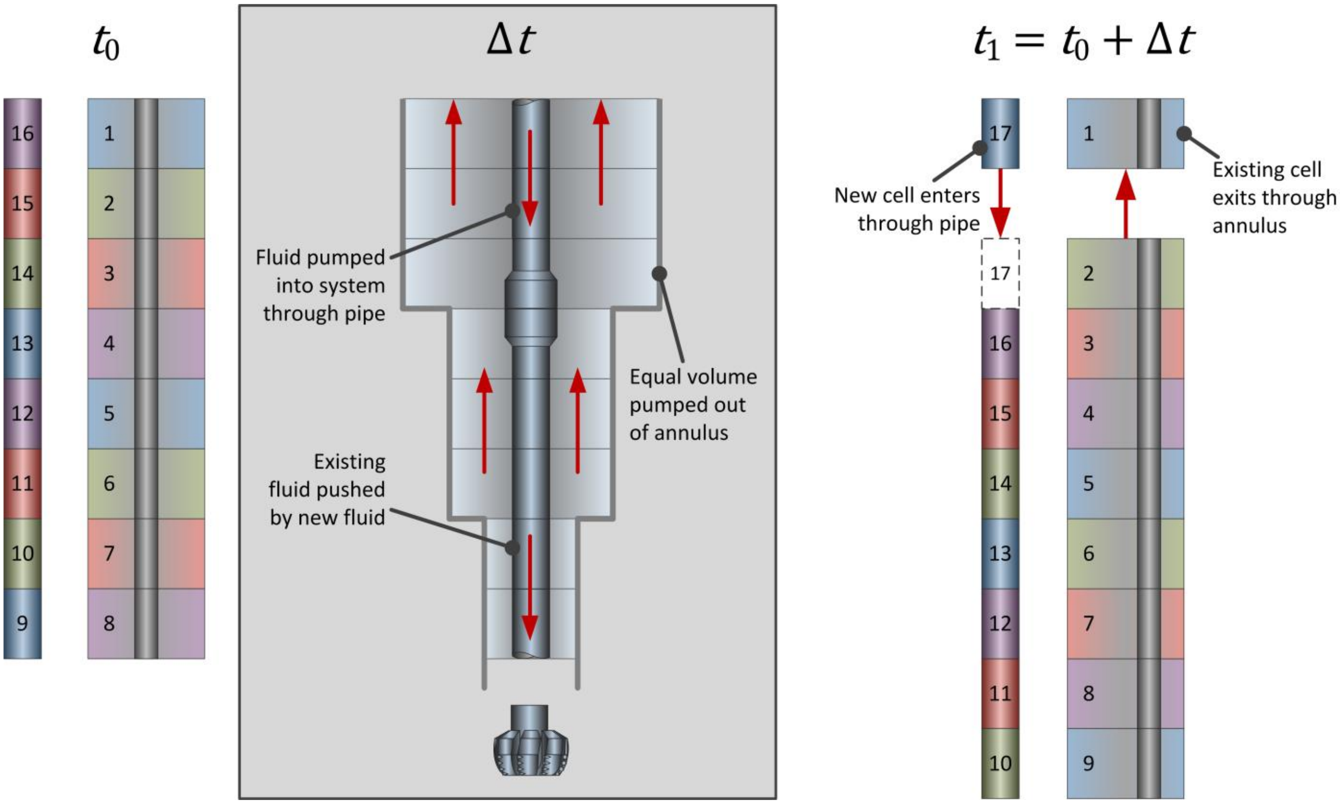
If the cells are too small, it may be necessary to merge some of them. The quality of the wellbore representation increases with additional cells, but the slower calculation speed hinders performance. Merging cells can achieve an optimal balance between speed and quality.



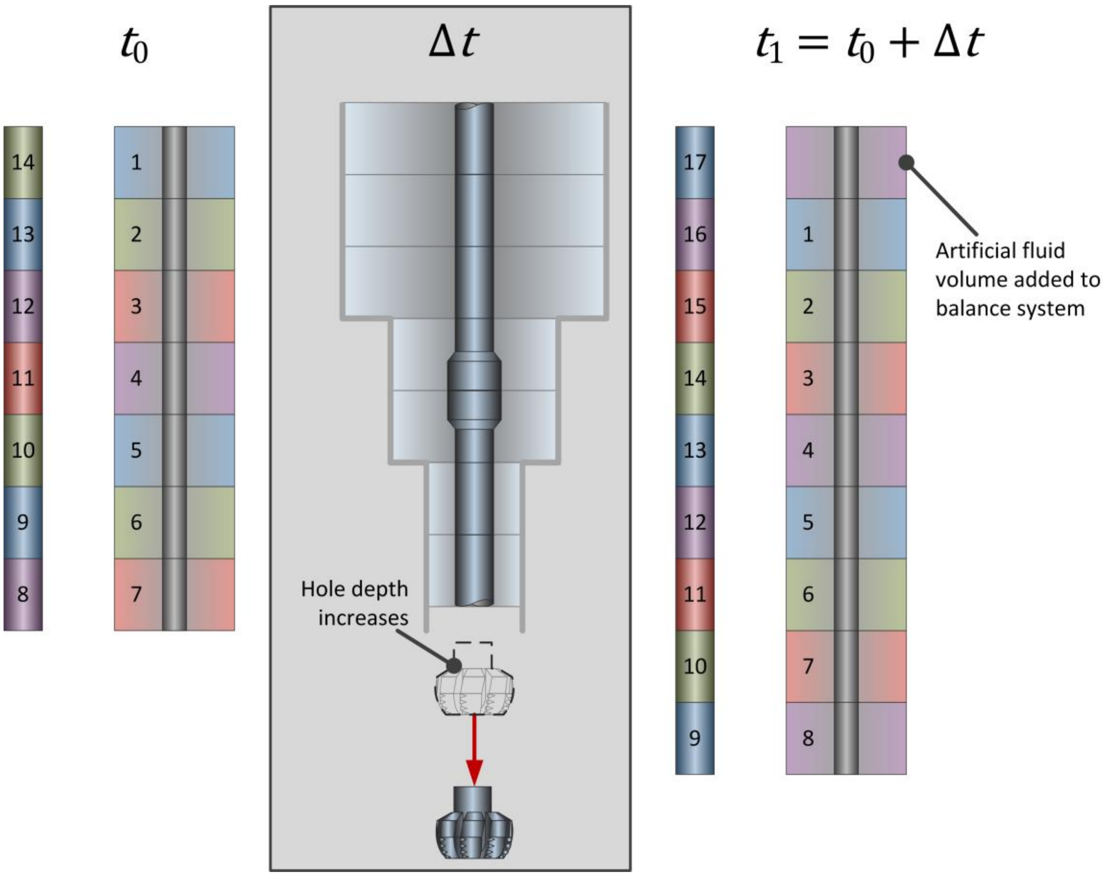
2.3.6 Operational Control Module

The operational control module simulates wellbore operations, such as circulation, tripping, and drilling. Wellbore conditions change depending on the wellbore operation, which is the driving force of the simulation. The operational control module stores the global operational properties, such as flow rate and bit depth, and uses them to control cell behavior by telling the displacement module how to move the cells.

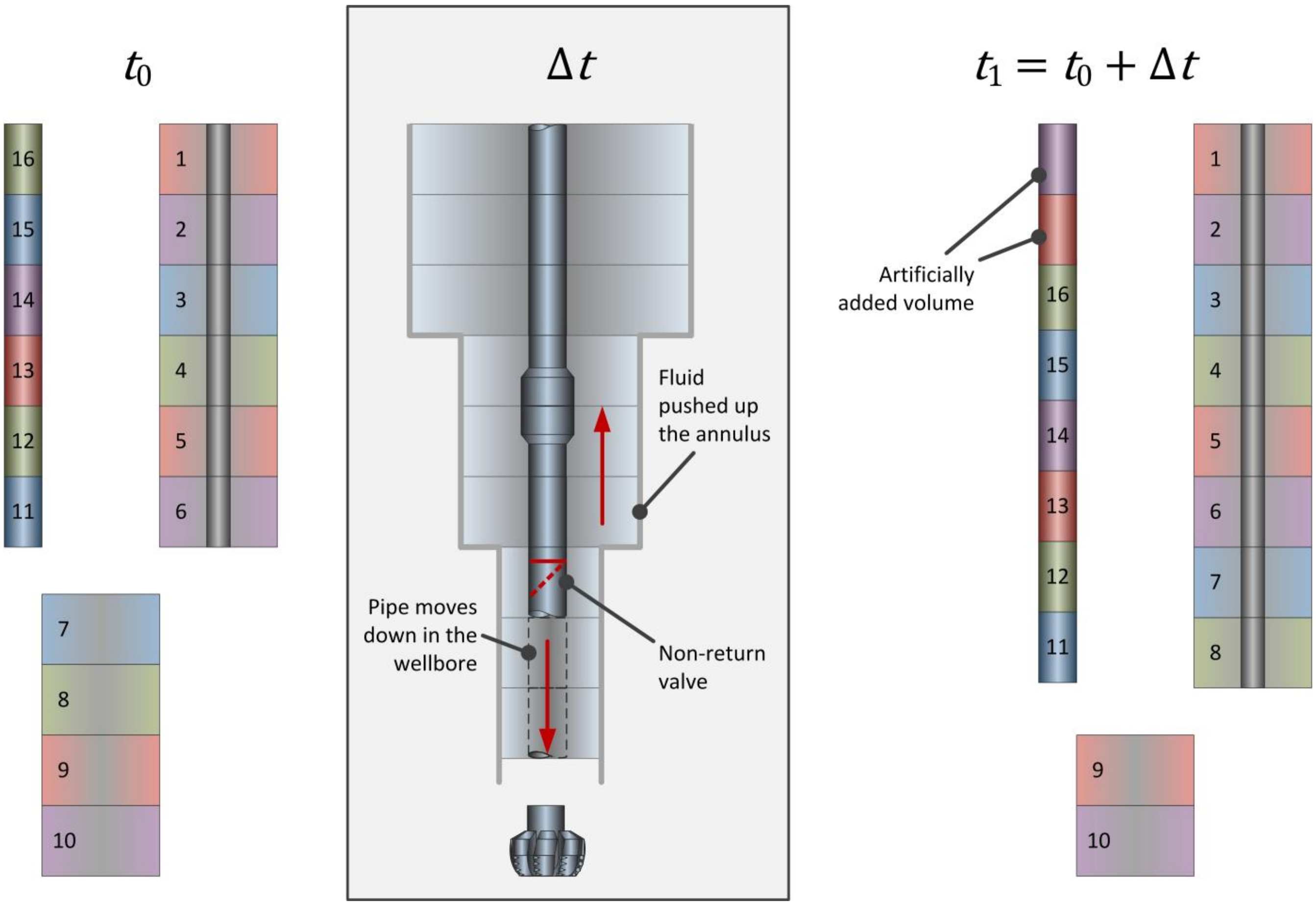
During circulation, new volume is added to the top of the list through the drillstring. Existing cells are pushed down by the new volume. The drillstring doesn't move during circulation, so the total volume in the system and the inactive fluid region below the bit do not change. An equal amount of volume exits the system at the end of the list through the annulus. Circulation may occur alone, such as when removing excess cuttings, or it may occur during drilling and tripping operations.



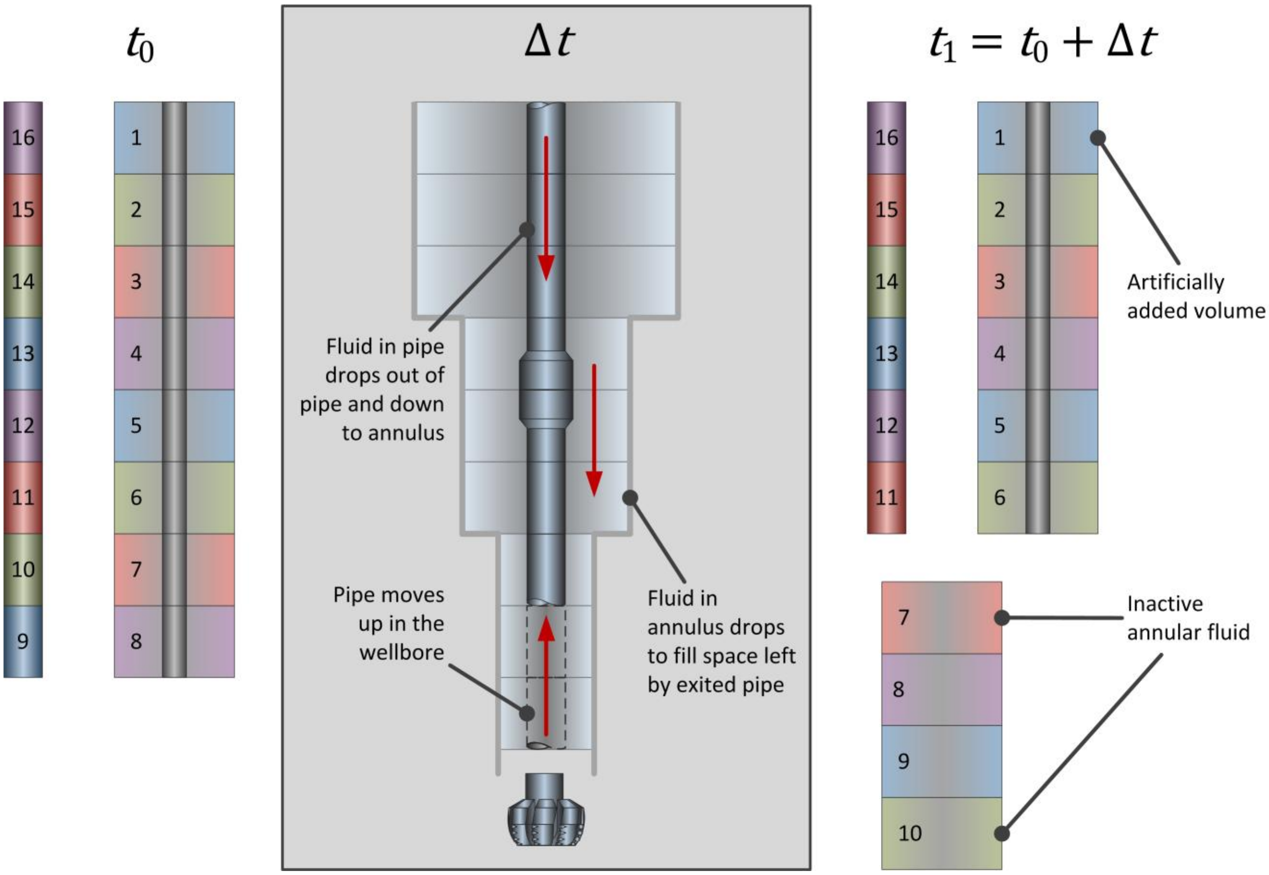
If the drillstring moves, the system volume is adjusted accordingly. Drilling adds volume to the system, and cells are moved to adjust for the change in pipe volume. The drill bit is on the ground, so there isn't an inactive fluid region below the bit. Volume may be artificially added to ensure fluid is present through the entire system.



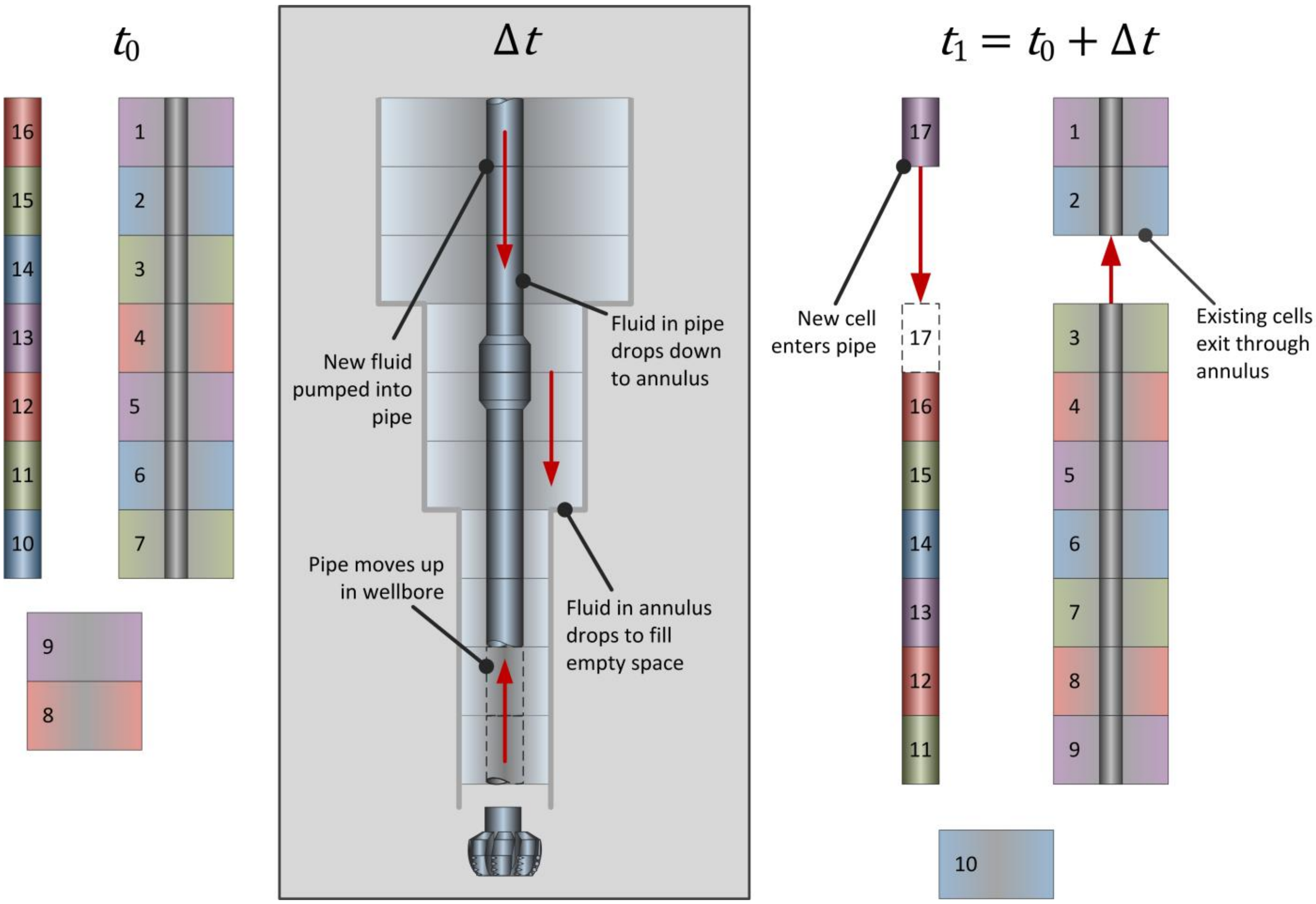
When tripping in without circulating, the bit depth increases. The pipe volume decreases because there is no fluid in the new pipe. The trip tank adds fluid to fill the new pipe region, and the inactive fluid below the bit is pushed into the annulus.



Similarly, tripping out without circulation causes fluid from the pipe to flow into the inactive region below the bit. The numerical framework assumes a non-return valve during tripping operations, which prevents the fluid from flowing back up the pipe.



However, fluid volume may be lost or gained when tripping out and circulating at the same time. Although fluid volume is lost when a pipe section exits the wellbore, circulation adds new volume. The amount of fluid gained or lost depends on the speed of the tripping operation and the flow rate of the fluid. When the flow rate is high and the trip speed is low, circulation replaces the fluid faster than the pipe is removed. When the flow rate is low and the trip speed is high, the fluid is exiting the system faster than circulation replaces it.



When tripping in while circulating, there are no gaps in volume because the fluid volume is displaced by the volume of the pipe.

2.3.7 Formulas Module

When the integration module performs a calculation, it accesses the equations stored in the formulas module. These equations include numerical methods, empirical relationships, and formulas for basic hydraulics calculations, such as pressure loss across simple and jet elements.

2.3.8 Integration Module

The integration module performs advanced calculations across the cells, controls the order of the calculations, and determines calculated properties. It selects the appropriate equation from the formulas module, uses the relevant input properties to calculate the correct value, and stores the resulting calculated property in the cell. Global calculated properties are sent to the post-processing module. When necessary, the integration module reconciles local calculated properties from cells with the global calculated properties of the system.

2.3.9 Post-Processing Module

The post-processing module stores global calculated properties for creating output properties and gathers local properties for generating profiles. At each iteration, new profiles are created to reflect cell properties.

Profiles are reported on a fixed grid, using results from the reference frame's moving grid. However, this causes some of the reporting points to have unknown values. The post-processing module estimates these values using linear interpolation. Some calculations also have unstable or chaotic behavior that causes oscillation and extraneous results in profiles. The post-processing module removes unnecessary data points to smooth profile graphs and show the overall trend, rather than the chatter from the calculated results.

Chapter 3: Isothermal Pressure Model

The conservation equations of mass, momentum, and energy describe the behavior of fluid. Assumptions and constraints reduce these equations into solvable formulas. However, certain aspects of fluid behavior can't be calculated when assumptions and constraints are used. Mathematical models mimic these lost fluid behaviors.

The isothermal pressure model assumes the drilling fluid is a single-phase fluid. By treating the individual fluid components as liquids, the need for multiphase representation is eliminated. The fluid density is also assumed to be constant.

The impact of temperature on fluid density and viscosity is also disregarded. Unsteady flow behavior is ignored, restricting fluid flow to steady, fully-developed flow across the pipe and annulus. Entrance flow is also ignored. To restrict fluid flow to one direction, the isothermal model assumes a non-return valve is present during tripping operations. A non-return valve prevents the fluid from flowing back into the drillstring.

Mathematical models simulate scenarios that are excluded by the assumptions and constraints placed on the conservation equations. The models used in the isothermal pressure model are listed below.

- Pressure loss due to turbulent and transitional flow regimes
- Pressure loss due to pipe rotation
- Additional pressure loss due to swab and surge
- Constitutive rheological models
- Bit and jet pressure loss
- Pressure loss due to contraction and expansion
- Pressure drops across complex geometries

3.1 Properties

The numerical framework uses input, output, and calculated properties. A property is a quantitative value that represents some aspect related to the drilling process.

- An *input* is a property sent to the simulator
- An *output* is any property produced by the simulator
- A *calculated property* is an intermediate property that is derived by the simulator from other properties.

Properties may be local or global properties. Local properties, such as pipe ID or cell volume, only apply to a particular cell. Global properties, such as flow rate or bit depth, apply to the entire simulation.

3.1.1 Input Properties

Geometry, fluid density, rheology, and the wellbore operation each affect pressure. To accurately calculate pressure, these factors must be taken into account.

Pressure calculations require inputs which are specific to the wellbore environment. Many wells are directionally drilled and have a curved trajectory, which impacts the distribution of hydrostatic pressure. Frictional pressure loss is affected by the type of geometry housing the fluid, the fluid's rheology, and the wellbore operation.

3.1.2 Geometry

The discretization module uses geometric input properties to discretize the wellbore and map local properties to the discretized cells

DRILLSTRING SEGMENTS

The geometric dimensions of an element are captured by defining the inside diameter, outside diameter, and length of the element. This information identifies the size and shape of the physical space occupied by the cell.

Property	Units	Definition
Pipe OD	in	Outside diameter of the pipe
Pipe ID	in	Inside diameter of the pipe
Length	ft	Length of the drillstring segment
Tool joint OD	in	Outside diameter of the tool joint
Tool joint ID	in	Inside diameter of the tool joint
Tool joint length	ft	Length of the tool joint
Surface pipe ID	in	Inside diameter of the pipe at the surface
Surface pipe length	ft	Length of the pipe at the surface

BIT

Due to its unique shape and function in the drillstring, the bit requires unique input properties. This information is used for bit-specific calculations, such as pressure loss across the bit.

Property	Units	Definition
TFA	in2	Total flow area
Nozzle diameter	1/32"	Diameter of the bit nozzles
K_d	--	Discharge efficiency coefficient

WELL INTERVALS

Well intervals are captured by defining the measured depth at the beginning and end of the interval. This identifies the vertical location of the cell in the wellbore.

Property	Units	Definition
Hole OD	in	Outside diameter of the casing, liner, or riser.
Hole ID	in	Inside diameter of the casing, liner, or riser.
Start depth	ft	Starting depth of the well interval.
End depth	ft	Ending depth of the well interval.

DEPTH

Depth is captured by defining the measured depth and TVD at specific points in the wellbore. This information is cross-referenced with the well interval data to identify the location of a cell.

Property	Units	Definition
Depth	ft	Measured depth. The distance from a point at the surface of the well, along the curve of the wellbore, to a specific point in the wellbore.
TVD	ft	True vertical depth. Vertical distance from a point at the surface of the well to a specific point in the wellbore.
Hole depth	ft	Measured depth at the bottom of the hole.
Bottomhole TVD	ft	True vertical depth at the bottom of the hole.

3.1.3 Fluid

The isothermal pressure model relies on fluid density and rheological properties to calculate pressure loss. Density, commonly referred to as mud weight, measures the mass of the fluid in an element. Only one value is needed for density in isothermal pressure calculations.

The values for rheological properties can typically be found on the mud report. They may be reported as Fann viscometer readings, PV and YP values, or LSYP values. These values are used to determine calculated properties, such as shear rate and flow regime.

Property	Units	Definition
Mud weight	lb/gal	Density of the fluid
R_{600}	rpm	Dial reading taken at 600 rpm
R_{300}	rpm	Dial reading taken at 300 rpm
R_{200}	rpm	Dial reading taken at 200 rpm
R_{100}	rpm	Dial reading taken at 100 rpm
R_6	rpm	Dial reading taken at 6 rpm
R_3	rpm	Dial reading taken at 3 rpm
n	none	Fluid behavior index, Herschel-Bulkley model
K	cP $lb^n/100\ ft^2$	Consistency index, Herschel-Bulkley model
	$lb/100\ ft^2$	Yield stress, Herschel-Bulkey model
PV	cP	Plastic viscosity, Bingham plastic model
YP	$lb/100\ ft^2$	Yield point, Bingham plastic model
$LSYP$	$lb/100\ ft^2$	Low shear yield point, Bingham plastic model

3.1.4 Operations

Operating parameters define the conditions of the simulated wellbore operation. They are used to determine calculated properties and for pressure calculations.

Property	Units	Definition
Flow rate	gal/min	Velocity of the fluid
Mud weight	lb/gal	Density of the fluid
Temperature	F°	Temperature of the fluid
Rotary	rpm	Speed at which the bit is rotating
Bit depth	ft	Depth of the bit
SBP	psi	Surface backpressure
ROP	ft/hr	Rate of penetration
Displacement index	--	Identifies a fluid being displaced into the wellbore

3.1.5 Cell Properties

Mapped, calculated, and time-persistent properties are stored in each cell in the system. These properties are used in global outputs and profiles.

MAPPED PROPERTIES

These properties are mapped to each cell.

Property	Units	Definition
Depth	ft	Measured depth of the top of a cell
TVD	ft	Measured TVD of the top of a cell
Volume	bbl	Total volume occupied by a cell
Cumulative volume at top	bbl	Total volume of all cells above a cell
Density	lb/gal	Fluid density of a cell
Fluid index	--	Unique fluid index that points to fluid properties
Length	ft	Measured length of a cell
TVD length	ft	Vertical length of a cell
OD	in	Outside diameter of a cell
ID	in	Inside diameter of a cell

CALCULATED PROPERTIES

These properties are calculated by the simulator and stored in the cell.

Property	Units	Definition
Hydraulics diameter	in	Equivalent circular diameter of duct-like geometry
Hydraulics area	in^2	Cross-sectional area defined using hydraulic diameter
Average velocity	ft/min	Average velocity of the fluid in a cell

Property	Units	Definition
Frictional pressure loss	psi	Frictional pressure loss across a cell
Hydrostatic pressure loss	psi	Hydrostatic pressure loss across a cell
Total pressure loss	psi	Total pressure loss across a cell
Friction factor	--	Skin friction factor at a cell for Fanning's equation
ECD	lb/gal	Equivalent circulating density of a cell
Reynold's number	--	Reynold's number of the fluid in the cell
Flow regime	--	Flow regime for a cell
PV	cP	Plastic viscosity of the fluid in a cell
YP	lb/100 ft^2	Yield point of the fluid in a cell
LSYP	lb/100 ft^2	Low shear yield point of the fluid in the cell

TIME-PERSISTENT PROPERTIES

These properties persist between timesteps.

Property	Units	Definition
Volume	bbl	Total volume occupied by a cell
Fluid index	--	Unique fluid index that points to fluid properties

3.1.6 Output Properties

The numerical framework produces global output properties, which apply to the entire system, and profile outputs, which are collections of local properties.

GLOBAL OUTPUT PROPERTIES

The overall values for pressure loss are stored as global output properties. They are single numerical values that represent the total pressure loss in a region.

Property	Units	Definition
PFann	psi	Total frictional pressure loss in the annulus
PHann	psi	Total hydrostatic pressure loss in the annulus
PTann	psi	Sum of the frictional and hydrostatic pressure loss in the annulus.
Pbit	psi	Pressure loss due to complex tools
SBP	psi	Surface backpressure
PFpipe	psi	Total frictional pressure loss in the pipe
PHpipe	psi	Total hydrostatic pressure loss in the pipe
PTpipe	psi	Sum of the frictional and hydrostatic pressure loss in the pipe
SPP	psi	Standpipe pressure
ECD	psi	Equivalent circulating density
Swab and surge	psi	Pressure loss due to swab and surge effects

PROFILE OUTPUTS

Profiles track unique property changes throughout the wellbore. The simulator creates profiles by comparing the cell properties listed earlier against the measured depth of each cell.

Profile	Units	Definition
Annular pressure loss	psi	Total pressure loss throughout the annulus
Pipe pressure loss	psi	Total pressure loss throughout the pipe
Total pressure loss	psi	Total pressure loss throughout the system
Annular velocity	ft/min	Fluid velocity throughout the annulus
Critical velocity	ft/min	Critical velocity points throughout the system
Flow regime	--	Flow regime types throughout the system
Reynold's number	--	Reynold's numbers throughout the system
Annulus volume	bbl	Cell volume throughout the annulus
Drillstring volume	bbl	Cell volume throughout the drillstring
Inactive volume	bbl	Inactive volume throughout the system
Fluid positions	--	Position of segments of fluid
Density	lb/gal	Fluid density throughout the system
PV	cP	Plastic viscosity values throughout the system
YP	lb/in^2	Yield point values throughout the system
LSYP	lb/in^2	Low shear yield point values throughout the system

3.2 Mathematical Models

The numerical framework uses mathematical models to determine the values for calculated properties, such as viscosity and average velocity. These formulas are designed to match Secure's results for pressure loss across simple and jet elements.

3.2.1 Geometry

Simple cells are defined by the volume they hold. Volume is calculated from the element's length and relevant diameters.

Eq. 3.1 Volume of a simple element, pipe

Eq. 3.2 Volume of a simple element, annulus

TVD and depth values are used to determine cell positions when the wellbore is discretized (Bourgoyne 1986).

Eq. 3.3 Change in TVD across a simple element

Eq. 3.4 Length of a simple element

Unit Conversion

Value	Type	Convert From	To	Multiply By
Volume	Input	gal	bbl	1/42

3.2.2 Hydrostatic Pressure

The principal factors affecting hydrostatic pressure are the true vertical depth of the cell and the density of the fluid that it contains.

Pressure Loss Across a Cell

Hydrostatic pressure loss across a cell is calculated from the cell's TVD and fluid density (API-13D 2006).

Eq. 3.5 Hydrostatic pressure loss across a simple element, pipe

Eq. 3.6 Hydrostatic pressure loss across a simple element, annulus

Integrating Pressure Loss Across a Region

Hydrostatic pressure loss for a region is calculated from the sum of the pressure loss values for all of the cells in an annular or pipe region (API-13D 2006).

Eq. 3.7 Hydrostatic pressure loss across a region, pipe

Eq. 3.8 Hydrostatic pressure loss across a region, annulus

These regional values are used to calculate the hydrostatic differential pressure loss. Hydrostatic differential pressure loss is the difference between hydrostatic pressure loss in the annulus and hydrostatic pressure loss in the pipe (Bourgoyne 1986).

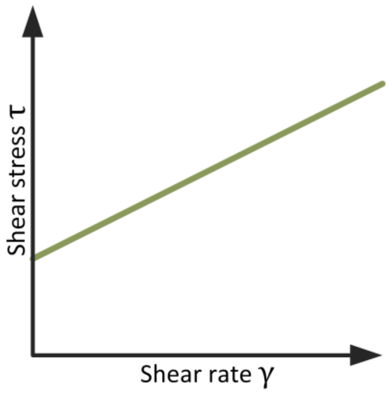
Eq. 3.9 Hydrostatic differential pressure loss

3.2.3 Frictional Pressure Loss

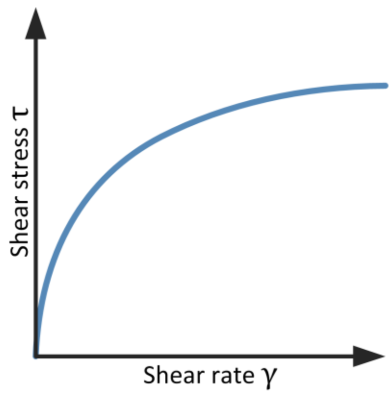
The numerical framework uses formulas that can match Secure's results for the Herschel-Bulkley, Power Law, and Bingham Plastic models. Because Secure appears to use a hybrid of techniques from multiple sources, it was necessary to use multiple sets of frictional pressure loss formulas to match the results generated by Secure.

Bingham plastic fluids will not flow until the yield shear stress is exceeded. After this point is exceeded, shear stress and shear rate have a linear relationship. Power law fluids, on the other hand, do not have a yield point and the relationship between shear stress and shear rate is not linear.

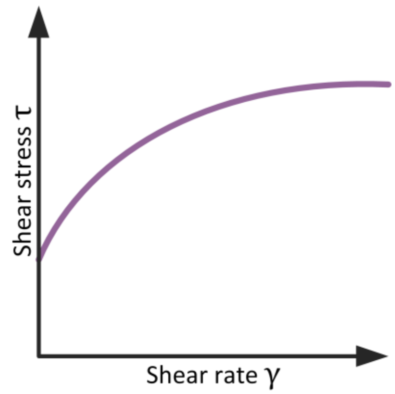
Calculations for Bingham plastic fluids are dependent on the shear rate and the yield point, while calculations for Power law fluids require values for the shear rate and flow behavior index (Bourgoyne 1986). The Herschel-Bulkley model, however, can be used to represent both of these types of fluids (Merlo 1995).



Eq. 3.10 Bingham plastic rheology model



Eq. 3.11 Power law rheology model



Eq. 3.12 Herschel-Bulkley rheology model

PARAMETERS

The numerical framework uses the formulas below to calculate values such as shear rate and shear stress from viscometer readings. Viscometer readings may be taken at several speeds, but not all of these values appear reliably on well reports. Although Fann data is the best way to characterize a fluid's rheology, many mud reports in the field do not provide enough data to use the Herschel-Bulkley model. Instead, mud reports often contain values for R_{600} and R_{300}, or for plastic viscosity and yield stress. The formulas below use readings taken at 300 rpm and 600 rpm, which are the most commonly reported readings (Bourgoyne 1986).

Eq. 3.13 RPM to Shear Rate

Eq. 3.14 Plastic viscosity

Eq. 3.15 Power law exponent

Eq. 3.16 Consistency index

Dial readings can also be used to calculate shear stress and yield stress (API-13D 2006).

Eq. 3.17 Fann dial to shear stress

Eq. 3.18 Yield Point

Flow rate can be calculated from the fluid's velocity and the physical dimensions of the cell (Merlo 1995).

Eq. 3.19 Flow rate, pipe

Eq. 3.20 Flow rate, annulus

Eq. 3.21 Fluid velocity, pipe

Eq. 3.22 Fluid velocity, annulus

The flow rate is also used to calculate average velocity (Bourgoyne 1986).

Eq. 3.23 Average velocity, pipe

Eq. 3.24 Average velocity, annulus

Apparent viscosity, the fluid's viscosity measured at a fixed shear rate, is required for the Herschel-Bulkley and Power Law models. A yield stress coefficient is used to account for fluids that have a yield stress (Merlo 1995).

Eq. 3.25 Apparent viscosity, pipe

Eq. 3.26 Apparent viscosity, annulus

Eq. 3.27 Correction factor, pipe

Eq. 3.28 Correction factor, annulus

For the Herschel-Bulkley and Power Law models, it is also necessary to use a geometric correction factor for the annulus.

Eq. 3.29 Geometric correction factor

HERSCHEL-BULKLEY MODEL

Since the Power law model cannot be used for a Bingham plastic fluid or vice versa, a hydraulics engine with only these two rheological models would require the user to select the appropriate rheological model before any calculations can be performed. However, using the Herschel-Bulkley model eliminates this problem because it can represent both types of fluid (Merlo 1995).

Unit Conversion

The formulas in the Herschel-Bulkley model use SI units, rather than standard oilfield units. Input values must be converted into SI units before any calculations are performed, and the resulting values need to be converted from SI units to standard oilfield units.

Value	Type	Convert From	To	Multiply By
Diameter	Input	in	m	0.3048/12.0
Radius	Input			
Length	Input	ft	m	0.3048
Flow rate	Input	gal/min	m^3/s	0.00378541178/60.0
τ_y	Input	$lb/100\ ft^2$	P	0.47880
k	Input	$lb/100\ ft^2$	P	0.47880
Density	Input	lb/gal	kg/m^3	119.8264
Pressure	Output	P	lb/in^2	1.45038 E — 04

Curve Fitting for Fann Data

Viscometer readings record the shear stress of a fluid at a fixed shear rate. While the Bingham plastic and Power law models only require readings taken at 300 rpm and 600 rpm, the Herschel-Bulkley model needs additional readings to determine the appropriate τ_y, k, and n values. Fann data, which consists of six viscometer readings at different shear rates, is used in the Herschel-Bulkley model. Because there are three unknown parameters, the readings are fitted to a curve to find the most reasonable values for τ_y, k, and n. To do this, a least squares problem is used for constrained optimization.

,

such that , where

Eq. 3.30 Curve fitting Fann data to the Herschel-Bulkley Model

The numerical method used to find the Herschel-Bulkley coefficients is the limited memory Broyden-Fletcher-Goldfarb-Shanno method, using the error term below.

Eq. 3.31 Error term used for Herschel-Bulkley coefficients

Flow Regime

Fluid flow can be described as laminar, transitional, or turbulent. Two dimensionless numbers, Reynold's number and Hedstrom's number, are used to calculate the characteristics of the fluid's flow regime. Reynold's number is used to determine if the flow regime is laminar, transitional, or turbulent.

Eq. 3.32 Reynold's number, pipe

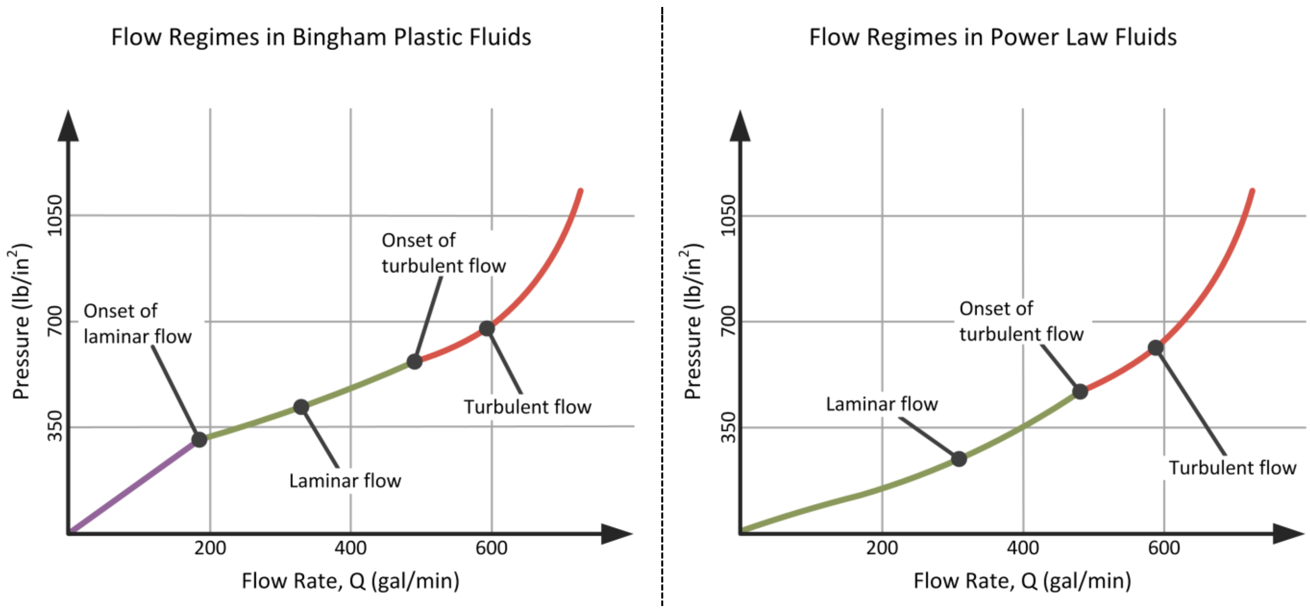
Eq. 3.33 Reynold's number, annulus

Hedstrom's number is used to determine the critical Reynold's numbers where changes in the flow regime occur.

Eq. 3.34 Hedstrom's number, pipe

Eq. 3.35 Hedstrom's number, annulus

Two critical points must be identified. The first is the Reynold's number at which the fluid begins to circulate and enters a laminar flow regime. If the fluid does not exceed the critical Reynold's number for laminar flow, the fluid will not flow. The second critical point occurs when the flow regime changes from laminar to turbulent.



The equation below is used to calculate the critical Reynold's number for laminar flow.

, where

Eq. 3.36 Critical Reynold's number for laminar flow, pipe and annulus

The critical flow rate required for flow to begin can be determined from critical Reynolds by using a numerical root finding method. The secant method is used in the prototype. A numerical method is required because an explicit formula could not be found.

The point at which flow changes from laminar to turbulent is defined by the critical Reynold's number for turbulent flow. A Reynold's number below this value indicates laminar flow, while a larger value indicates turbulent flow (Merlo 1995).

Eq. 3.37 Critical Reynold's number for turbulent flow, pipe

Eq. 3.38 Critical Reynold's number for turbulent flow, annulus

Pressure Loss Across A Simple Element

The friction factor, f, is the last component needed to calculate the frictional pressure loss across a simple element. It is calculated from the Reynold's number (Merlo 1995).

Eq. 3.39 Friction factor for laminar flow, pipe

Eq. 3.40 Friction factor for laminar flow, annulus

Eq. 3.41 Friction factor for turbulent flow, pipe

Eq. 3.42 Friction factor for turbulent flow, annulus

Once the appropriate friction factor has been determined, frictional pressure loss can be calculated.

Eq. 3.43 Frictional pressure loss for laminar flow, pipe

Eq. 3.44 Frictional pressure loss for laminar flow, annulus

Eq. 3.45 Frictional pressure loss for turbulent flow, pipe

Eq. 3.46 Frictional pressure loss for turbulent flow, annulus

Eq. 3.47 Frictional pressure loss before circulation begins, pipe

Eq. 3.48 Frictional pressure loss before circulation begins, annulus

POWER LAW MODEL

The Power Law model is one of the more conventional rheological models. Although the Herschel-Bulkley model is the preferred approach, the numerical framework also contains the necessary formulas to calculate frictional pressure loss using this model. For the Power Law model, the numerical framework uses the Herschel-Bulkley model in the pipe (Merlo 1995) and a modified form of the Bingham Plastic model in the annulus (Bourgoyne 1986). For this reason, standard oilfield units must be converted into SI units for pipe calculations, as outlined earlier for the Herschel-Bulkley model.

Flow Regime

In the pipe, the Reynold's number is calculated using the yield stress coefficient defined earlier.

,

Eq. 3.49 Reynold's number, pipe

Eq. 3.50 Reynold's number, annulus

Eq. 3.51 Hedstrom's number, pipe

Eq. 3.52 Hedstrom's number, annulus

The only critical Reynold's number for Power law fluids is the point where the flow regime changes from laminar to turbulent. Because Power law fluids do not have a yield stress, there is no critical point where laminar flow begins.

Eq. 3.53 Critical Reynold's number for turbulent flow, pipe

Eq. 3.54 Critical Reynold's number for turbulent flow, annulus

Pressure Loss Across A Simple Element

An explicit friction factor is needed to calculate pressure loss in the pipe. However, the friction factor in the annulus does not need to be calculated explicitly because the annular pressure loss equation already accounts for it.

Eq. 3.55 Friction factor for laminar flow, pipe

Eq. 3.56 Friction factor for turbulent flow, pipe

Once the friction factor has been determined, frictional pressure loss can be calculated.

Eq. 3.57 Frictional pressure loss for laminar flow, pipe

Eq. 3.58 Frictional pressure loss for laminar flow, annulus

Eq. 3.59 Frictional pressure loss for turbulent flow, pipe

Eq. 3.60 Frictional pressure loss for turbulent flow, annulus

The Dodge-Metzner correlation is also used to account for turbulent flow (Bourgoyne 1986).

Eq. 3.61 Dodge-Metzner correlation

BINGHAM PLASTIC MODEL

The numerical framework uses a modified form of the Bingham Plastic model in the pipe and the annulus (Bourgoyne 1986).

Flow Regime

Unlike the Herschel-Bulkley and Power Law models, a yield stress coefficient is not required when determining Reynold's number.

Eq. 3.62 Reynold's number, pipe

Eq. 3.63 Reynold's number, annulus

Eq. 3.64 Hedstrom's number, pipe

Eq. 3.65 Hedstrom's number, annulus

Hedstrom's number is used to determine the critical Reynold's number for the onset of laminar flow. The value for Hedstrom's number determines which of the formulas below should be used to calculate this critical Reynold's number.

,

Eq. 3.66 Critical Reynold's number for laminar flow, pipe and annulus

The critical Reynold's number for the onset of laminar flow is used to find the corresponding critical flow rate needed for flow to begin.

Eq. 3.67 Critical flow rate, pipe

Eq. 3.68 Critical flow rate, annulus

The point at which the flow regime changes from laminar to turbulent flow is also calculated from Hedstrom's number. This is a modified version of Hank's formula (Bourgoyne 1986).

,

where , , and

Eq. 3.69 Critical Reynold's number for turbulent flow, pipe and annulus

Pressure Loss Across A Simple Element

When the flow regime is laminar, an explicit friction factor does not need to be calculated because it is already accounted for in the pressure loss equation.

Eq. 3.70 Frictional pressure loss for laminar flow, pipe

Eq. 3.71 Frictional pressure loss for laminar flow, annulus

Eq. 3.72 Frictional pressure loss before circulation begins, pipe

Eq. 3.73 Frictional pressure loss before circulation begins, annulus

However, when the flow regime is turbulent, the Blasius correlation is used to account for the additional pressure loss caused by turbulent flow. The same friction factor formula is used in the pipe and the annulus (Bourgoyne 1986).

Eq. 3.74 Blasius correlation

Once the friction factor has been determined, frictional pressure loss can be calculated for turbulent flow regimes.

Eq. 3.75 Frictional pressure loss for turbulent flow, pipe

Eq. 3.76 Frictional pressure loss for turbulent flow, annulus

DUE TO PIPE ROTATION

Additional frictional pressure is lost due to the rotation of the pipe. This factor below is used to correct the correlation for pipe rotation when the average velocity falls below a critical limit (Hemphill 2008).

, where

Eq. 3.77 Frictional pressure loss due to pipe rotation

INTEGRATING FRICTIONAL PRESSURE

The frictional pressure loss for each region is found by adding the values for pressure loss across all of the cells in that region (API-13D 2006).

Eq. 3.78 Integrated frictional pressure loss, pipe

Eq. 3.79 Integrated frictional pressure loss, annulus

The frictional pressure loss throughout the wellbore can be found from the sums of the total pipe and annular frictional pressure losses.

Eq. 3.80 Integrated frictional pressure loss, wellbore

3.2.4 Other Pressure Losses

Frictional and hydrostatic pressure losses are not sufficient to give a complete picture of the various pressures at work in the wellbore. Pressure loss at the bit, standpipe pressure, and bottomhole pressure must also be calculated for adequate analysis of the hydraulics system.

BIT PRESSURE LOSS

Pressure loss at the bit is based on a kinetic energy change that occurs as the fluid is pushed through the nozzles of the drill bit, and is dependent on the volume of fluid flowing through the bit nozzles. The total fluid area (TFA) can be calculated from the diameters of the bit nozzles, which are often expressed in 32nds of an inch (API-13D 2006).

Eq. 3.81 Conversion of bit nozzle diameters to TFA

The discharge coefficient, K_d, is used to account for the frictional pressure loss across the bit nozzles (Warren 1989). The value of this coefficient varies according to the type of drill bit because K_d is dependent on the ratio between the output and input diameters of the nozzles.

- For PDC bits, the value of K_d is 0.97
- For roller cone bits, the value of K_d is 1.03

Pressure loss at the bit can be calculated with the same formula, regardless of rheological model, because the effects of viscosity are negligible when a fluid flows through short nozzles.

Eq. 3.82 Pressure loss at the bit

STANDPIPE PRESSURE

The pressure in the standpipe, which transports the fluid from the pumps to the drillstring, is found from the surface backpressure, frictional and hydrostatic pressure losses and bit pressure (API-13D 2006).

Eq. 3.83 Standpipe Pressure

BOTTOMHOLE PRESSURE

It is not currently possible to directly measure the pressure at the bottom of the borehole. However, bottomhole pressure can be found from the sum of the integrated values for hydrostatic and frictional pressure losses in the annulus (API-13D 2006).

Eq. 3.84 Bottomhole Pressure

Appendix A: Nomenclature

Symbol	Definition	Standard Units	SI Units
	Total flow area	in^2	mm^2
	Correction factor for Reynold's number for an annular element	--	--
	Correction factor for Reynold's number for a pipe element	--	--
	Annulus outside diameter	in	mm
	Pipe inside diameter	in	mm
	Pipe outside diameter	in	mm
	Measured depth	ft	m
	Friction factor	--	--
	Friction factor in the annulus	--	--
	Friction factor in the pipe	--	--
	Friction factor for laminar flow in the annulus	--	--
	Friction factor for laminar flow in the pipe	--	--
	Friction factor for turbulent flow in the annulus	--	--
	Friction factor for turbulent flow in the pipe	--	--
	Geometric correction factor	--	--
	Hedstrom's number in the annulus	--	--
	Hedstrom's number in the pipe	--	--
	Represents the ith cell in a range	--	--
	Consistency index	$lb^n/100\ ft^2$	Pa^n
	Discharge coefficient	--	--
	Denotes laminar flow	--	--

Symbol	Definition	Standard Units	SI Units
	Length	ft	m
	Length of an annular element	ft	m
	Length of a pipe element	ft	m
	Power-law exponent	--	--
	Number of elements	--	--
	Bottomhole pressure loss	lb/in^2	kPa
	Pressure loss at the bit	lb/in^2	kPa
	Frictional pressure loss	lb/in^2	kPa
	Frictional pressure loss for an annular element	lb/in^2	kPa
	Frictional pressure in a pipe element	lb/in^2	kPa
	Frictional pressure loss due to pipe rotation	lb/in^2	kPa
	Hydrostatic pressure loss	lb/in^2	kPa
	Hydrostatic pressure loss in an annular element	lb/in^2	kPa
	Hydrostatic pressure loss in a pipe element	lb/in^2	kPa
	Surface backpressure loss	lb/in^2	kPa
	Standpipe pressure loss	lb/in^2	kPa
	Flow rate in the annulus	gal/min	dm^3/s
	Flow rate at the bit	gal/min	dm^3/s
	Flow rate in the pipe	gal/min	dm^3/s
	Critical flow rate required for flow to begin in an annular element	gal/min	dm^3/s
	Critical flow rate required for flow to begin in a pipe element	gal/min	dm^3/s
	Annulus outside radius	in	mm

Symbol	Definition	Standard Units	SI Units
	Drillpipe inside radius	in	mm
	Drillpipe outside radius	in	mm
	Viscometer reading at 3 rpm	° deflection	° deflection
	Viscometer reading at 6 rpm	° deflection	° deflection
	Viscometer reading at 100 rpm	° deflection	° deflection
	Viscometer reading at 200 rpm	° deflection	° deflection
	Viscometer reading at 300 rpm	° deflection	° deflection
	Viscometer reading at 600 rpm	° deflection	° deflection
	Reynold's number for an annular element	--	--
	Reynold's number for a pipe element	--	--
	Critical Reynold's number for laminar flow	--	--
	Critical Reynold's number for laminar flow in an annular element	--	--
	Critical Reynold's number for laminar flow in a pipe element	--	--
	Critical Reynold's number for turbulent flow in an annular element	--	--
	Critical Reynold's number for turbulent flow in a pipe element	--	--
	Critical Reynold's number for turbulent flow	--	--
	Denotes turbulent flow	--	--
	True vertical depth	ft	m
	True vertical depth of an annular element	ft	m
	True vertical depth of a pipe element	ft	m
	Velocity of the fluid in annular element	ft/min	m/s
	Average velocity of fluid in the annulus	ft/min	m/s

Symbol	Definition	Standard Units	SI Units
	Velocity of the fluid in the pipe	ft/min	m/s
	Average velocity in a pipe element	ft/min	m/s
	Volume of annulus element	bbl	m^3
	Volume of a pipe element	bbl	m^3
	Shear rate	s^{-1}	s^{-1}
	Shear stress that corresponds to a R_{300} viscometer reading.	lbf/100 ft^2	Pa
	Shear stress that corresponds to a R_{600} viscometer reading.	lbf/100 ft^2	Pa
	Shear stress that corresponds to a given viscometer reading.	lbf/100 ft^2	Pa
	Apparent viscosity of a fluid in the annulus	eq cP	Pa•s^n
	Apparent viscosity of a fluid in the pipe	eq cP	Pa•s^n
	Plastic viscosity	cP	Pa•s
	Plastic viscosity in an annular element	cP	Pa•s
	Plastic viscosity in a pipe element	cP	Pa•s
	Density of the fluid in an annular element	lb/gal (ppg)	kg/m^3
	Density of the fluid at the bit	lb/gal (ppg)	kg/m^3
	Density of the fluid in a pipe element	lb/gal (ppg)	kg/m^3
	Shear stress	lbf/100 ft^2	Pa
	Shear stress reading	lbf/100 ft^2	Pa
	Initial shear stress	lbf/100 ft^2	Pa
	Rotary pipe rotation	rpm	rpm

Appendix B: References

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